

## Experiment no:5

Experiment Name: image for log tranmission

code:

```
In [ ]: import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

img_path = 'input.jpg'
img = cv2.imread(img_path)

if img is None:
    print("Error: Image not found.")
else:

    img_float = img.astype(np.float32)
    c = 255 / np.log(1 + np.max(img_float))
    log_img = c * (np.log(1 + img_float))
    log_img = np.clip(log_img, 0, 255).astype(np.uint8)

    plt.figure(figsize=(12, 6))

    plt.subplot(1, 2, 1)
    plt.title("Original Image")
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.axis('off')

    plt.subplot(1, 2, 2)
    plt.title("Log Transformed")
    plt.imshow(cv2.cvtColor(log_img, cv2.COLOR_BGR2RGB))
    plt.axis('off')

    folder = "image for Log-Transformation"
    if not os.path.exists(folder): os.makedirs(folder)

    output_path = os.path.join(folder, 'comparison_result.png')
    plt.savefig(output_path)
    plt.show()

    print(f"Comparison saved as {output_path}")
```

Original Image



Log Transformed



Comparison saved as image for Log-Transformation/comparison\_result.png